

Experiment : 06

Experiment Name: Familiarization with different display elements.

Objectives: The objectives of this experiment are:

- To learn about 7-segment display.
- To learn about alphanumeric display.
- To learn about LCD display.
- To learn about LED matrix, display.

Seven segment Display

The 7-segment display, also written as “seven segment display”, consists of seven LEDs (hence its name) arranged in a rectangular fashion as shown. Each of the seven LEDs is called a segment because when illuminated the segment forms part of a numerical digit (both Decimal and Hex) to be displayed. An additional 8th LED is sometimes used within the same package thus allowing the indication of a decimal point, (DP) when two or more 7-segment displays are connected together to display numbers greater than ten.

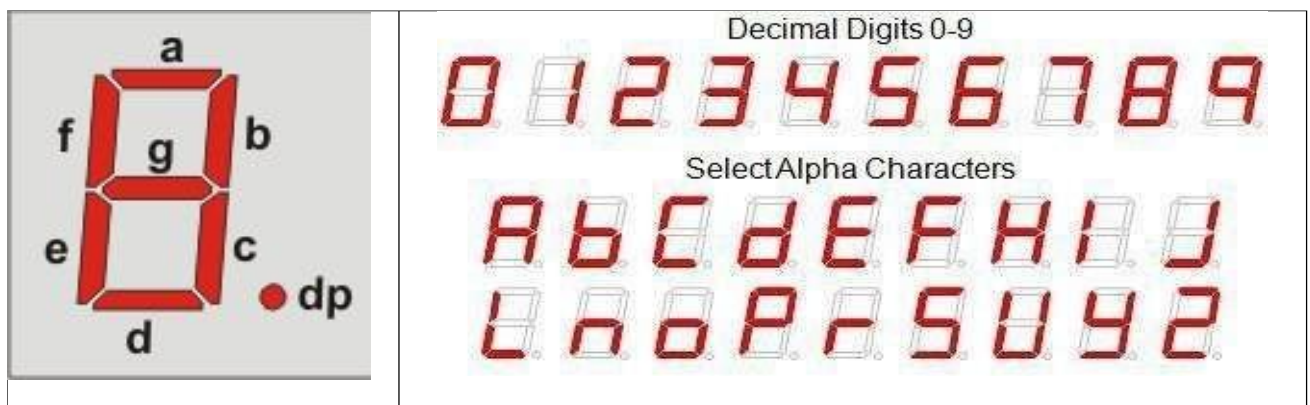


Fig 1: Seven segment display and pattern of digits and character
Types of 7-segment Display:

The displays common pin is generally used to identify which type of 7-segment display it is. As each LED has two connecting pins, one called the “Anode” and the other called the “Cathode”, there are therefore two types of LED 7-segment display called: Common Cathode (CC) and Common Anode (CA).

The difference between the two displays, as their name suggests, is that the common cathode has all the cathodes of the 7-segments connected directly together and the common anode has all the anodes of the 7-segments connected together and is illuminated as follows.

The Common Cathode (CC) – In the common cathode display, all the cathode connections of the LED segments are joined together to logic “0” or ground. The individual segments are illuminated by application of a “HIGH”, or logic “1” signal via a current limiting resistor to forward bias the individual Anode terminals (a-g).

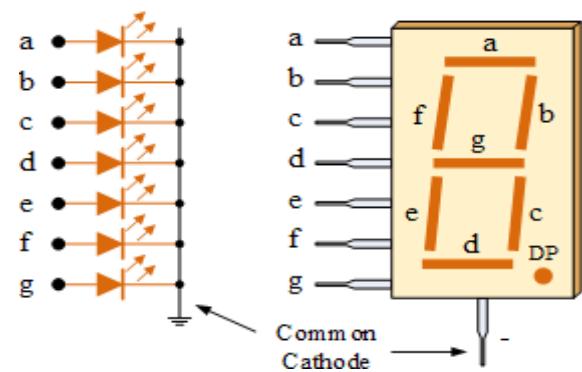


Fig 2: Common Cathode 7-segment display

The Common Anode (CA) – In the common anode display, all the anode connections of the LED segments are joined together to logic “1”. The individual segments are illuminated by applying a ground, logic “0” or “LOW” signal via a suitable current limiting resistor to the Cathode of the particular segment (a-g).

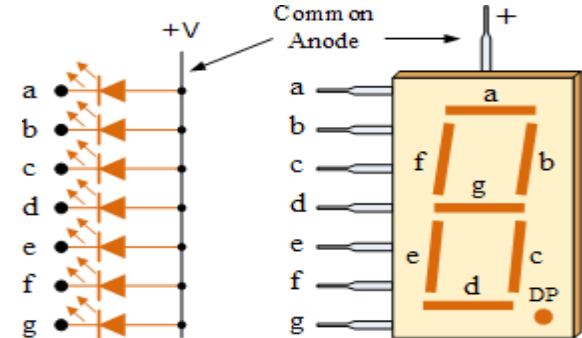


Fig 3: Common Anode 7-segment display.

In general, common anode displays are more popular as many logic circuits can sink more current than they can source.

Displaying in 7-segment display

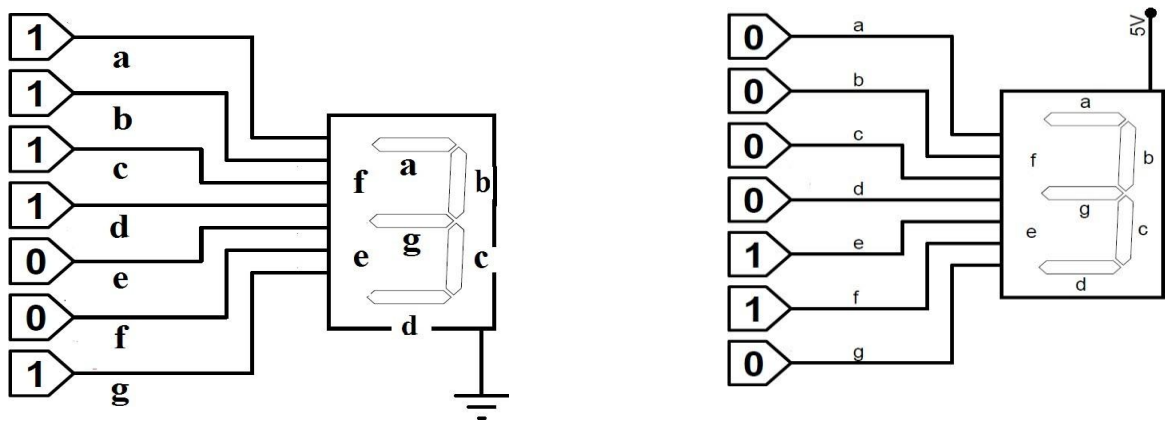


Fig 3.1 (a): Displaying 3 in common cathode 7-segment display **Fig 3.2 (b):** Displaying 3 in common anode 7-segment display

A BCD to 7-segment decoder can be used to convert BCD number into 7-segment digit pattern.

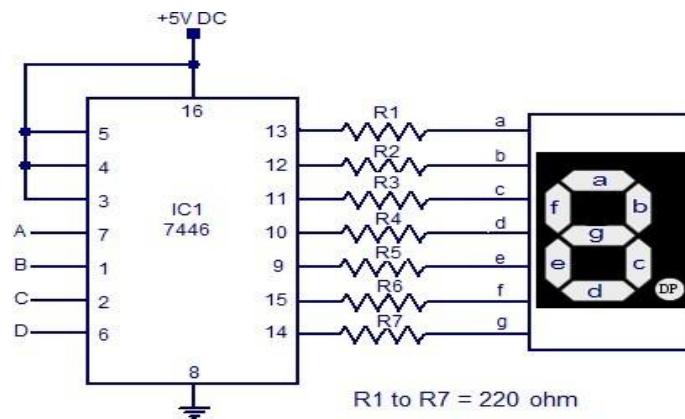


Fig 3.3: Use of 7-segment decoder

Displaying in multiple 7-segment display

Displays connected to the microcontroller usually occupy a large number of valuable I/O pins, which can be a big problem especially if it is needed to display multi digit numbers. The problem is more than obvious if, for example, it is needed to display two 6-digit numbers (a simple calculation shows that 96 output pins are needed in this case). The solution to this problem is called MULTIPLEXING. Only one digit is active at a time, but they change their state so quickly making impression that all digits of a number are simultaneously active.

First a byte representing units is applied on a microcontroller port and a transistor T1 is activated at the same time. After a while, the transistor T1 is turned off, a byte representing tens is applied on a port and a transistor T2 is activated. This process is being cyclically repeated at high speed for all digits and corresponding transistors.

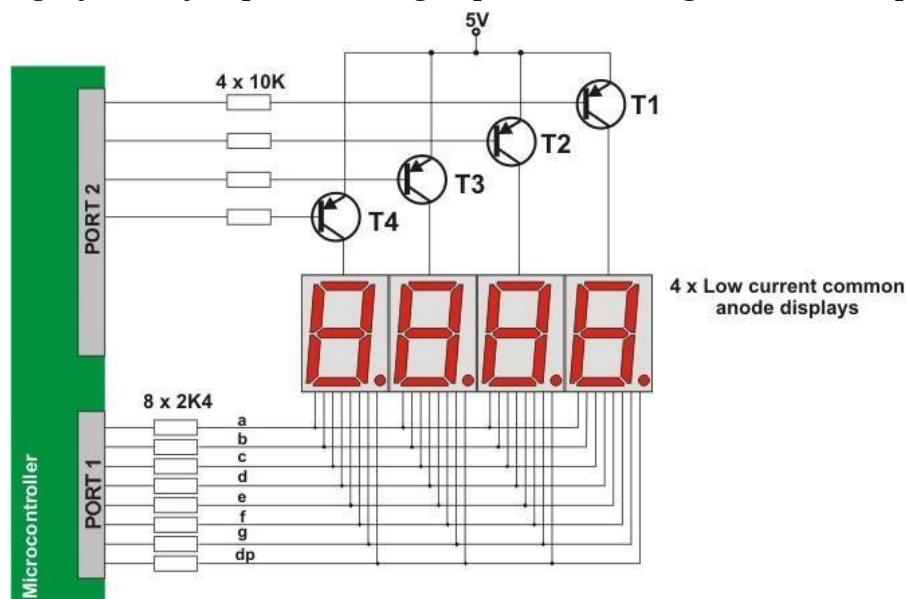


Fig 4: Multiplexing technique to show number in multiple seven segment display

Pin diagram of 7-segment display

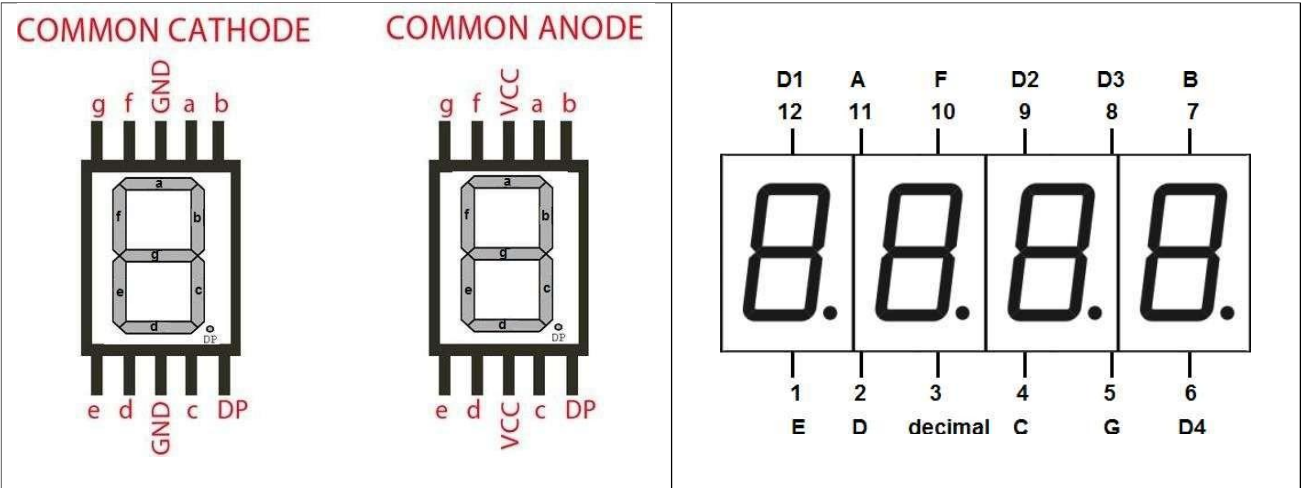
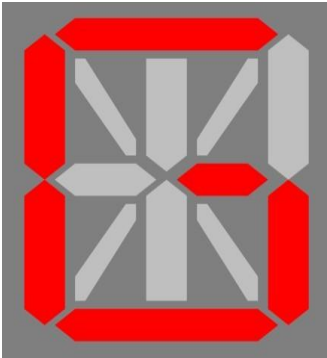
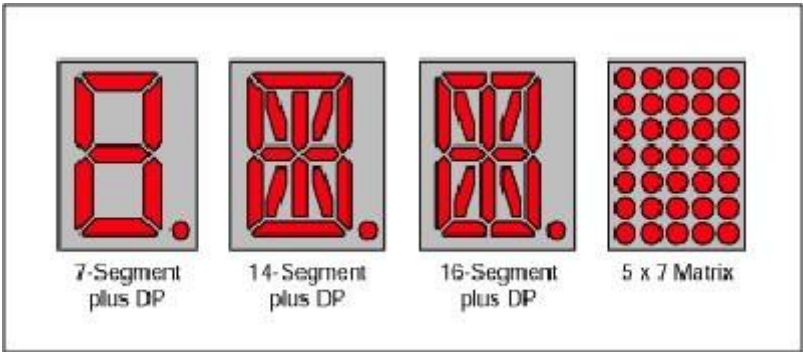


Fig 5: Pin diagram of 7-segment display

Alphanumeric display



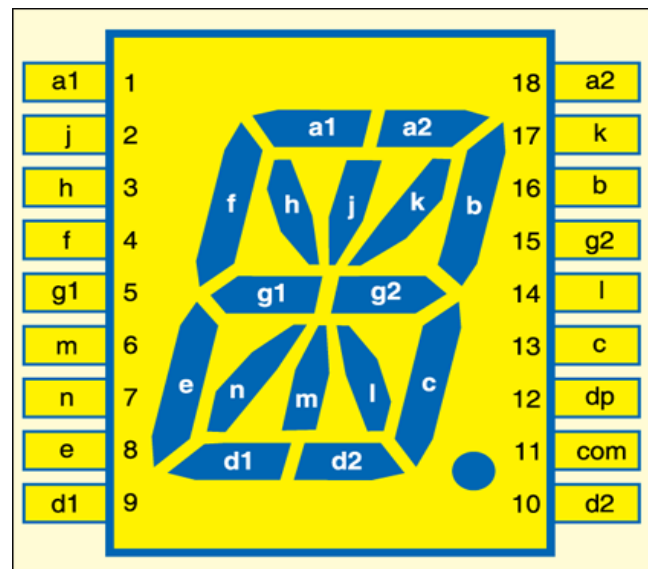


Fig. 6: Alphanumeric display

About LCD Display:

This component is specialized to be used with the microcontrollers, which means that it cannot be activated by standard IC circuits. It is used for displaying different messages on a miniature liquid crystal display. A model described here is for its low price and great capabilities most frequently used in practice. It is based on the HD44780 microcontroller (Hitachi) and can display messages in two lines with 16 characters each. It displays all letters of alphabet, greek letters, punctuation marks, mathematical symbols etc. In addition, it is possible to display symbols made up by the user. Other useful features include automatic message shift (left and right), cursor appearance, LED backlight etc.



Fig. 7: LCD Display

Along one side of a small printed board there are pins used for connecting to the microcontroller. There are in total of 14 pins marked with numbers (16 in case the backlight is built in). Their functions are described in table below:

Function	Pin Number	Name	Logic State	Description
Ground	1	Vss	-	0V
Power supply	2	Vdd	-	+5V
Contrast	3	Vee	-	0 - Vdd
Control of operating	4	RS	0	D0 – D7 are interpreted as commands
			1	D0 – D7 are interpreted as data
	5	R/W	0	Write data (from controller to LCD)
			1	Read data (from LCD to controller)
	6	E	0	Access to LCD disabled
			1	Normal operating
Data / commands			From 1 to 0	Data/commands are transferred to LCD
	7	D0	0/1	Bit 0 LSB
	8	D1	0/1	Bit 1
	9	D2	0/1	Bit 2
	10	D3	0/1	Bit 3
	11	D4	0/1	Bit 4
	12	D5	0/1	Bit 5
	13	D6	0/1	Bit 6
	14	D7	0/1	Bit 7 MSB

Table: Pin functions of 2X16 LCD display

LCD Screen

LCD screen consists of two lines with 16 characters each. Every character consists of 5x8 or 5x11 dot matrix. This book covers 5x8 character display, which is indeed the most commonly, used one.

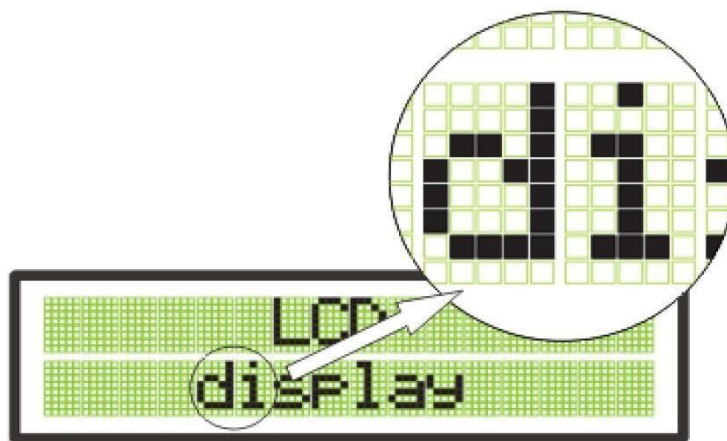


Fig 8: LCD screen.

Display contrast depends on power supply voltage. For that reason, varying voltage 0-Vdd is applied on the pin marked as Vee. Trimmer potentiometer is usually used for that purpose. Some LCD displays have built in backlight (blue or green diodes). When used during operation, a current limiting resistor should be serially connected to one of the pins for backlight (similar to LED diodes).

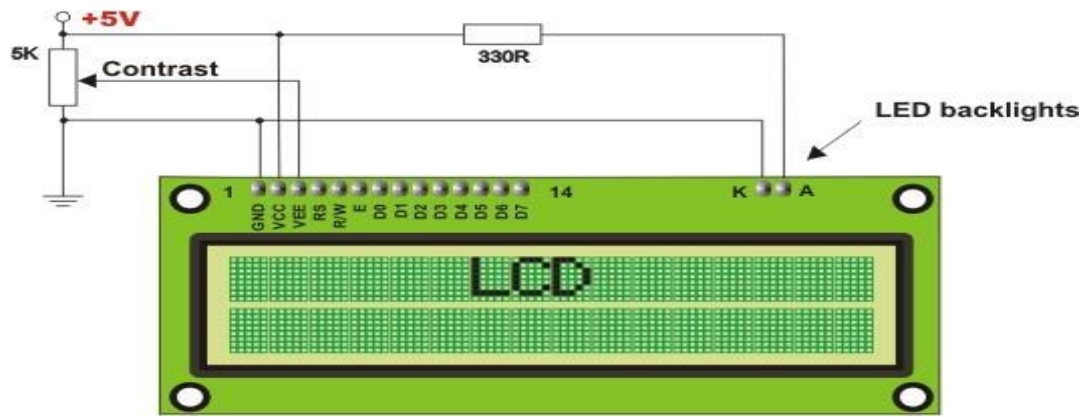


Fig 9: LCD contrast adjustment.

LCD Memory

LCD display contains three memory blocks:

DDRAM-Display

Data RAM CGRAM-

Character Generator

RAM CGROM-

Character Generator

ROM

DDRAM Memory: DDRAM memory is used for storing characters that should be displayed. The size of this memory is sufficient for storing 80 characters. Some memory locations are directly connected to the characters on display.

All works quite simply: it is enough to configure display to increment addresses automatically (shift right) and set starting address for the message that should be displayed (for example 00 hex).

After that, all characters sent through lines D0-D7 will be displayed as a message. In this very case, displaying starts from the first field of the first line because the address is 00 hex. If more than 16 characters are sent then all of them will be memorized, but only first sixteen characters will be visible. In order to display the rest of them, a shift command should be used. Virtually, everything looks as if LCD display is a window which shifts left-right over memory locations containing different characters. In reality, that is how the effect of message shifting on the screen has been made.

DDRAM Memory

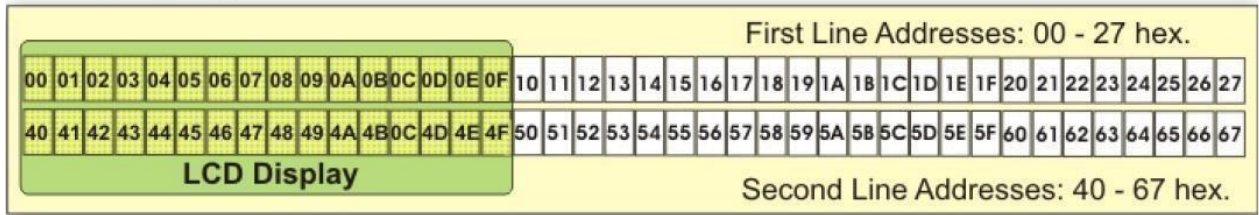


Fig 10: memory address of DDRAM memory

If cursor is on, it appears at location which is currently addressed. In other words, when a character appears at cursor position, it will automatically move to the next addressed location. This is a sort of RAM memory so data can be written to and read from it, but its contents is irretrievably lost upon the power goes off.

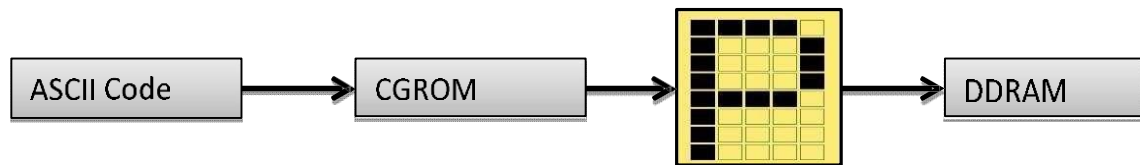
CGROM Memory: CGROM memory contains default character map with all characters that can be displayed on the screen. Each character is assigned to one memory location.

Table 2.1: CGROM memory address for different character

Lower Addr		Upper Addr	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111		
CG RAM LOCATIONS FOR 8 CUSTOM CHARACTERS	xxxx0000	0000	0000			0	1	A	P	^	P				一	タ	ミ	α	p	
	xxxx0001	0001	0001			!	1	A	Q	a	q				。	ア	チ	△	ä	q
	xxxx0010	0010	0010			"	2	B	R	b	r				「	イ	ツ	×	β	θ
	xxxx0011	0011	0011			#	3	C	S	c	s				」	ウ	テ	モ	ε	ω
	xxxx0100	0100	0100			\$	4	D	T	d	t				、	エ	ト	ヤ	μ	Ω
	xxxx0101	0101	0101			%	5	E	U	e	u				・	オ	ナ	ユ	σ	Ü
	xxxx0110	0110	0110			&	6	F	V	f	v				ヲ	カ	ニ	ヨ	ρ	Σ
	xxxx0111	0111	0111			'	7	G	W	g	w				フ	キ	ヌ	ラ	q	π
CG ROM LOCATIONS CONTAINS PREDEFINED FONTS	xxxx1000	1000	1000			(	8	H	X	h	x				イ	ク	ネ	リ	フ	Σ
	xxxx1001	1001	1001			)	9	I	Y	i	y				ろ	ケ	ノ	ル	リ	Y
	xxxx1010	1010	1010			*	:	J	Z	j	z				エ	コ	ハ	レ	j	チ
	xxxx1011	1011	1011			+	;	K	L	k	l				オ	サ	ヒ	ロ	*	ス
	xxxx1100	1100	1100			,	<	L	¥	l	l				ヤ	シ	フ	ワ	Φ	円
	xxxx1101	1101	1101			-	=	M	J	m	>				ユ	ズ	ヘ	ン	も	÷
	xxxx1110	1110	1110			.	>	N	^	n	→				ヨ	セ	ホ	ン	ん	
	xxxx1111	1111	1111			/	?	O	_	o	←				ッ	ソ	マ	□	る	

Addresses of CGROM memory locations match the characters of ASCII. If the program being currently executed encounters a command “send character P to port” then binary value 0101 0000 appears on the port. This value is ASCII equivalent to the character P. It is further written to LCD, which results in displaying the symbol from the 0101 0000 location of CGROM. In other words, the character “P” is displayed. This applies to all letters of the alphabet

(capital and small), but not to the numbers!



Character Pattern

Fig 11: Process of displaying ASCII character.

As seen on the previous map, addresses of all digits are pushed forward by 48 in relative to their values (digit 0 address is 48, digit 1 address is 49, digit 2 address is 50 etc.). Accordingly, in order to display digits correctly it is necessary to add a decimal number 48 to each of them prior to sending them to LCD.

Modification of character map is not possible in CGROM.

CGRAM: This memory works same as CG ROM but as this is RAM we can modify its content any time. So, this is the place where when have to first store our custom character pattern, then that pattern can be sent to display.

The HD44780 has total 8 CG RAM memory locations. So, we can generate only up to 8 custom characters. But you can always change the content of CG RAM on the fly to generate new characters. The addresses of 8 CG RAM locations go from 0x00 to 0x07.

LCD Connection

Depending on how many lines are used for connecting LCD to the microcontroller, there are 8-bit and 4-bit LCD modes. The appropriate mode is selected at the beginning of the operation in that process called initialization. 8-bit LCD mode uses outputs D0-D7 to transfer data as explained previous.

The main purpose of 4-bit LED mode is to save valuable I/O pins of the microcontroller. Only 4 higher bits (D4-D7) are used for communication, while others may be unconnected. Each data is sent to LCD in two steps- four higher bits are sent first (normally through the lines D4-D7) and four lower bits are sent afterwards. Initialization enables LCD to link and interpret received bits correctly.

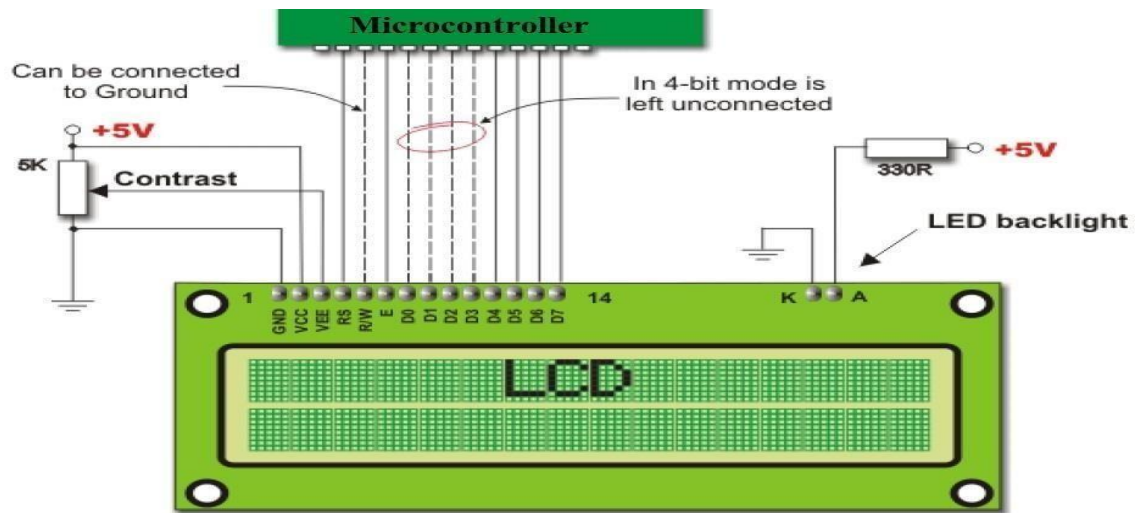


Fig 12: LCD connection with microcontroller.

Besides, data is rarely read from LCD (it is mainly transferred from the microcontroller to LCD) so it is often possible to save an extra I/O pin by simple connecting R/W pin to the Ground. Such saving has its price. Messages will be normally displayed, but it will not be possible to read busy flag since it is not possible to read display as well. Fortunately, there is a simple solution. After sending a character or a command it is important to give LCD enough time to do its job. Owing to the fact that execution of the slowest command lasts for approximately 1.64mS, it will be fairly enough to wait approximately 2mS for LCD.

Dot Matrix Display

In a dot matrix display, multiple LEDs are wired together in rows and columns. This is done to minimize the number of pins required to drive them. For example, a 8×8 matrix of LEDs (shown below) would need 64 I/O pins, one for each LED pixel. By wiring all the anodes together in rows (R1 through R8), and cathodes in columns (C1 through C8), the required number of I/O pins is reduced to 16. Each LED is addressed by its row and column number. In the figure below, if R4 is pulled high and C3 is pulled low, the LED in fourth row and third column will be turned on. Characters can be displayed by fast scanning of either rows or columns.

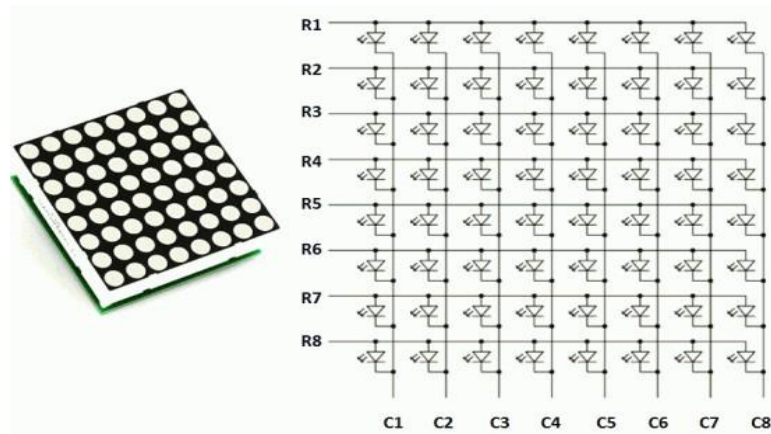
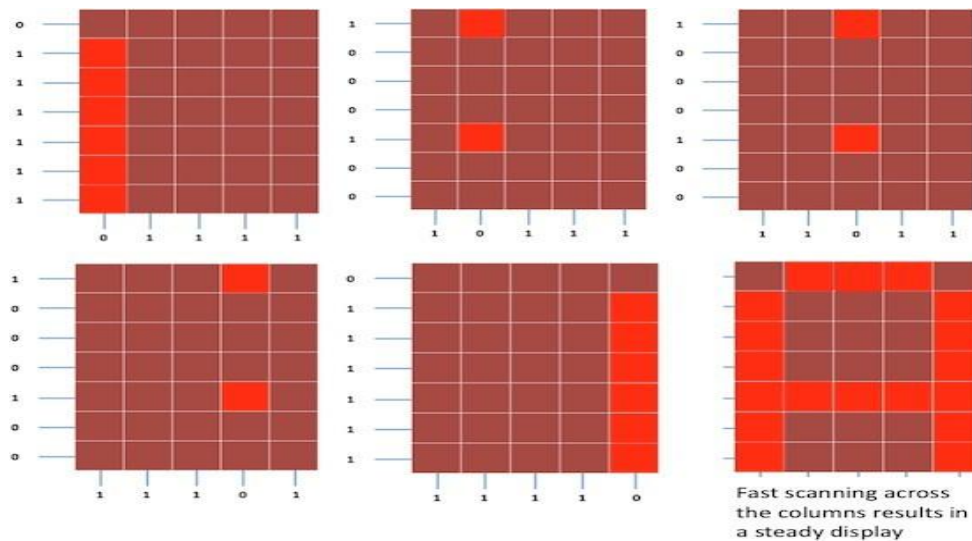


Fig 13: Dot Matrix Display

Now, we will learn how to display still characters by column scanning in a standard 5×7- pixel format. Suppose, we want to display the alphabet A. We will first select the column C1 (which means C1 is pulled low in this case), and deselect other columns by blocking their ground paths (one way of doing that is by pulling C2 through C5 pins to logic high). Now, the first column is active, and you need to turn on the LEDs in the rows R2 through R7 of this column, which can be done by applying forward bias voltages to these rows.



Next, select the column C2 (and deselect all other columns), and apply forward bias to R1 and R5, and so on. Therefore, by scanning across the column quickly (> 100 times per second), and turning on the respective LEDs in each row of that column, the persistence of vision comes in to play, and we perceive the display image as still.

Discussion:

In this lab, different display devices such as 7-segment, alphanumeric, LCD, and LED matrix displays were studied. The 7-segment display is simple and mainly used for showing numbers. Alphanumeric displays can show both letters and numbers, making them more useful for text display. LCD displays are low-power and can display characters and symbols clearly. LED matrix displays are suitable for scrolling text and patterns. This lab helped to understand the basic working and applications of different display devices.